

Viva Questions

1. What is Flutter?
2. What is StatefulWidget?
3. What is StatelessWidget?
4. What is `setState()`?
5. What is `ListView.builder`?
6. What is a `TextEditingController`?
7. What is the purpose of `Scaffold`?
8. What is the difference between Row and Column widgets?
9. Why is `setState()` required in Flutter?
10. What are the advantages of Flutter for mobile application development?

Viva Questions with Answers

Experiment 7: To-Do List Application

1. What is Flutter?

Answer: Flutter is an open-source UI toolkit developed by Google for building cross-platform mobile, web, and desktop applications using a single codebase.

2. What is Dart?

Answer: Dart is the programming language used by Flutter for developing applications.

3. What is a StatefulWidget?

Answer: A StatefulWidget is a widget whose data can change during the execution of the application.

4. What is a StatelessWidget?

Answer: A StatelessWidget is a widget whose data cannot change after it is created.

5. What is `setState()`?

Answer: `setState()` is used to update the UI whenever the state of a StatefulWidget changes.

6. Why is StatefulWidget used in the To-Do List application?

Answer: It is used because tasks are added and deleted dynamically, requiring the UI to update continuously.

7. What is ListView.builder?

Answer: `ListView.builder` is a widget used to create a scrollable list of items dynamically and efficiently.

8. What is TextEditingController?

Answer: It is used to read and manage the text entered in a TextField.

9. What is the purpose of the Scaffold widget?

Answer: Scaffold provides the basic structure of a Flutter screen, including AppBar, Body, Drawer, and Floating Action Button.

10. What happens when a task is deleted?

Answer: The task is removed from the list, and `setState()` refreshes the UI to display the updated list.

Experiment 8: Calculator Application

1. What is the purpose of this calculator application?

Answer: To perform basic arithmetic operations such as addition, subtraction, multiplication, and division.

2. Which widget is used to accept user input?

Answer: TextField widget.

3. Why is TextEditingController used?

Answer: To retrieve the values entered by the user in the TextField.

4. Which widget is used to display buttons?

Answer: ElevatedButton widget.

5. Why is StatefulWidget used in the calculator?

Answer: Because the result changes whenever the user performs an operation.

6. What is the use of setState() in the calculator application?

Answer: It updates the result displayed on the screen after performing calculations.

7. What are arithmetic operators?

Answer: Symbols used to perform mathematical operations such as:

- + Addition
- - Subtraction
- * Multiplication
- / Division

8. What is the purpose of the Row widget?

Answer: Row arranges widgets horizontally on the screen.

9. What is the purpose of the Column widget?

Answer: Column arranges widgets vertically on the screen.

10. What will happen if the user enters 10 and 5 and presses the "+" button?

Answer: The application displays:

Result: 15.0

1. What are widgets in Flutter?

Answer: Widgets are the basic building blocks of a Flutter application used to create the user interface.

2. What is Hot Reload?

Answer: Hot Reload allows developers to see code changes instantly without restarting the application.

3. What is the difference between Hot Reload and Hot Restart?

Answer:

- Hot Reload: Updates UI while preserving app state.
- Hot Restart: Restarts the application and resets the state.

4. What are the advantages of Flutter?

Answer:

- Single codebase
- Fast development
- Beautiful UI
- Cross-platform support
- High performance

5. What is MaterialApp?

Answer: MaterialApp is the root widget that provides Material Design functionality to a Flutter application.

6. What is an AppBar?

Answer: AppBar is a toolbar displayed at the top of the application screen.

7. What is the difference between MainAxisAlignment and CrossAxisAlignment?

Answer:

- MainAxisAlignment controls alignment along the main axis.
- CrossAxisAlignment controls alignment along the cross axis.

8. What is the purpose of Expanded widget?

Answer: Expanded allows a widget to occupy the available space inside a Row or Column.

9. What is Flutter SDK?

Answer: Flutter SDK is a software development kit containing tools, libraries, and frameworks required to build Flutter applications.

10. Why is Flutter popular?

Answer: Flutter is popular because it enables developers to build high-performance applications for multiple platforms using a single codebase.